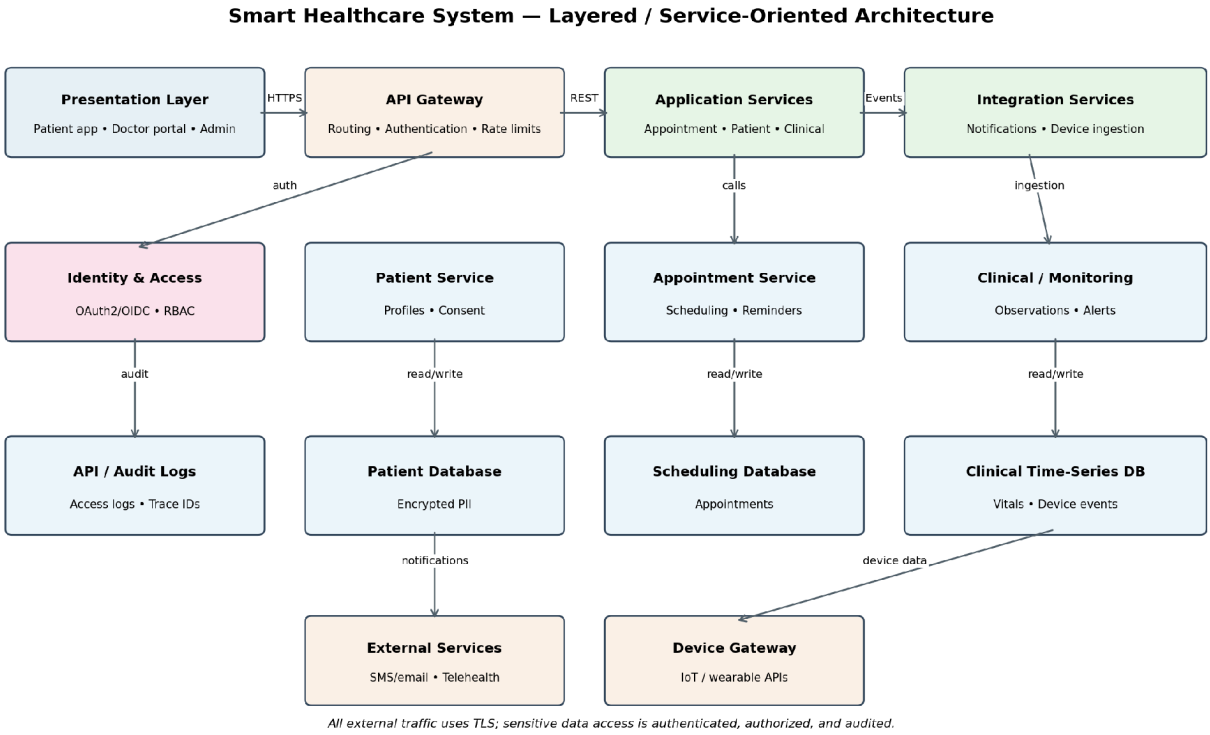
ASSIGNMENT 7

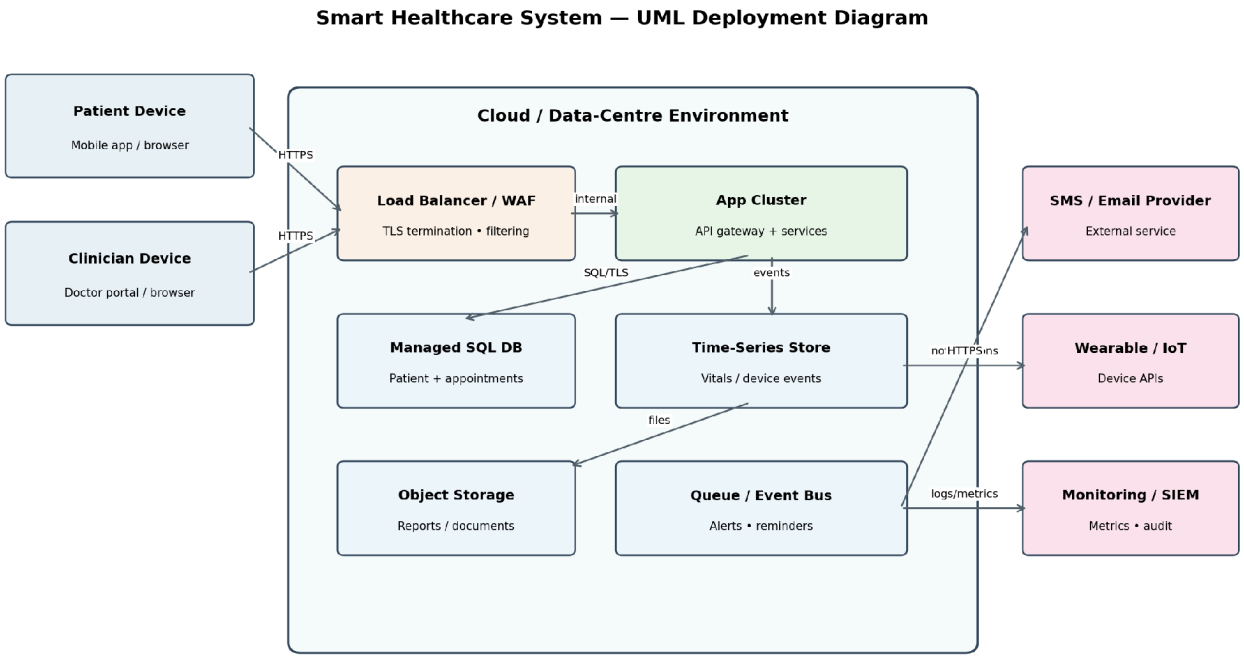
1. Proposed Architecture

The design uses a service-oriented architecture organized into presentation, API/access, application-service, data, and integration/observability responsibilities. The API gateway provides a single secure entry point. Patient, appointment, and clinical-monitoring services own their business logic and data stores. An event queue supports asynchronous notifications and alert processing.



2. Deployment Diagram

Patient and clinician devices access the platform over HTTPS. A load balancer/Web Application Firewall forwards traffic to a horizontally scalable application cluster. Managed SQL storage keeps patient and appointment records; a time-series store holds vital readings; object storage keeps reports; and an event bus decouples alerts and reminders. External SMS/email and wearable APIs are accessed through controlled integrations. Monitoring and SIEM collect metrics, traces, and audit events.



3. Brief Architecture Justification

Separation of concerns: The presentation, access, business-service, data, and integration layers have distinct responsibilities. This improves maintainability and makes testing easier.

Independent scalability: Appointment scheduling and clinical monitoring may have different workloads. Separate services allow capacity to be scaled where needed.

Security and privacy: TLS protects data in transit; OAuth 2.0/OpenID Connect and role-based access control restrict access. Consent checks, encryption at rest, least privilege, and audit logs help protect sensitive health information.

Reliability: Queues and asynchronous event processing prevent slow SMS/email providers from blocking core workflows. Retries, dead-letter queues, backups, and health checks improve resilience.

Data ownership and consistency: Each service owns its data and exposes controlled APIs/events. Cross-service workflows should use idempotent handlers and eventual consistency where appropriate.

Interoperability: A device gateway and API adapters isolate vendor-specific wearable protocols. Healthcare integrations can adopt standards such as HL7 FHIR where applicable.

Observability: Centralized logs, metrics, traces, request IDs, and security events help teams detect failures and investigate incidents.